Transfer-Learning Strategies for Big-5 Wildlife Classification

Engineering report: structural changes, performance shifts, and error analysis (ResNet50, TensorFlow, Colab GPU)

1. Summary

- **All three strategies land at 97.7% to 98.3% test accuracy** on a 299-image test set (feature extraction 97.99%, partial fine-tune 97.66%, full fine-tune 98.33%).
- **The differences are 1 to 2 images**, and the 95% Wilson intervals overlap almost completely (about 95.5% to 99.1%). On this evidence the three methods cannot be ranked with confidence.
- **Cost differs sharply.** Full fine-tuning trains about 1,900 times more parameters than the frozen-backbone head (23.49M vs 12,294) and takes about 1.9 times the wall time of Experiment A for its stage (313.7 s vs 167.9 s).
- **Fine-tuning fixed some errors and added one.** Leopard recall rose from 92.6% to 96.3% for full fine-tuning, but one rhino was newly misclassified as zebra.
- **Four class-level error patterns persist across all three models** (buffalo to elephant, buffalo to rhino, zebra to rhino, leopard to zebra). Reviewing the images suggests most come from multi-animal scenes, a small subject, and probable label noise, not from missing model capacity.
- **Main validity risk:** lion and leopard come only from the Kaggle source, and the other four classes only from Ultralytics, so species and image source are confounded. Section 6 covers this and other limitations.

2. Structural changes

2.1 Data pipeline

The pipeline changed in four ways between the first draft and the executed notebook. The first change corrected a labeling defect.

Change	Before	After	Why it matters
Kaggle labeling	Folder-name matching	Class id read from each YOLO label file, names from *data.yaml*	The Kaggle set is a 54-class detection dataset with no per-species folders. Folder matching had tagged frogs, moths and spiders as lion and leopard.
Class balance	Uncapped (thousands of lion/leopards)	Capped at 400 per class	Final counts: buffalo 374, elephant 374, rhino 371, zebra 385, lion 307, leopard 177.
Splits	Original dataset splits	All images pooled, then stratified 70/15/15 (1391 / 298 / 299)	Keeps six classes in every split; may leak near-duplicates (see section 6).
Imbalance handling	None	Class weights during training (leopard 1.87, lion 1.08, others about 0.88)	Leopard is the smallest class and gets the largest loss weight.

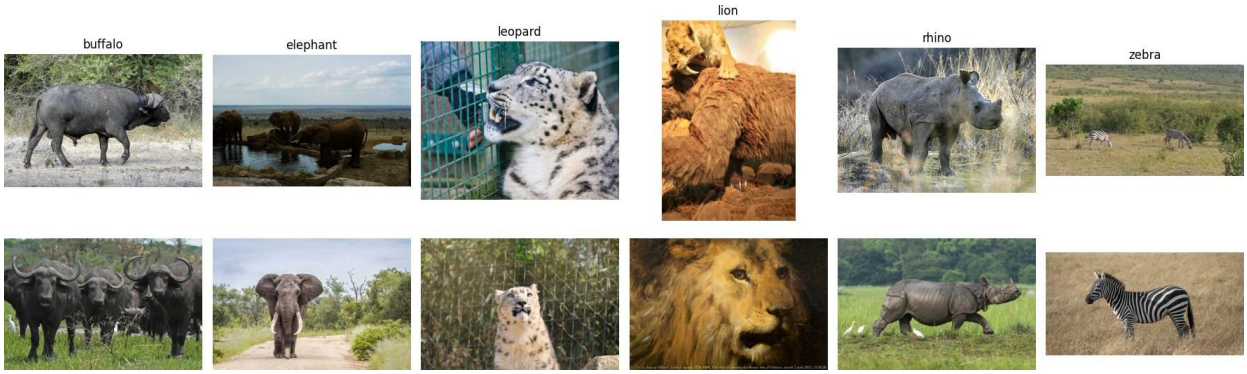


Figure 1. Two random samples per class after the labeling fix. Lion and leopard now show the correct animals.

2.2 Model architecture across experiments

All three experiments share one architecture: augmentation, ResNet50 preprocessing, ResNet50 backbone (ImageNet weights, called with *training=False*), global average pooling, dropout 0.3, and a 6-way softmax. Only what is trainable changes. B and C start from the head trained in A. BatchNorm layers stay frozen in every experiment.

	A. Feature extraction	B. Partial fine-tune	C. Full fine-tune
Trainable layers	Dense head only	conv5 stage + head	Whole backbone + head
Trainable parameters	12,294	14,965,766	23,493,766
Learning rate	1e-3	5e-5	1e-5
Epochs run (max 15)	15	11 (early stop; best epoch 7)	15 (best epoch 14)
Wall time	167.9 s	151.0 s (stage only)	313.7 s (stage only)
Steady-state time per epoch	about 9 s	about 11 s	about 18 s
Best validation loss	0.0557	0.0268	0.0387

Timing notes: A's time includes a 38 s first epoch (graph compilation). B and C times exclude A's head training, so the full recipe costs 319 s for B and 482 s for C in total.

3. Performance shifts

Experiment	Test accuracy	95% CI (Wilson)	Macro-F1	Errors / 299
A. Feature extraction	0.9799	95.7% - 99.1%	0.9786	6
B. Partial fine-tune	0.9766	95.3% - 98.9%	0.9738	7
C. Full fine-tune	0.9833	96.2% - 99.3%	0.9837	5

Class (test support)	A recall	B recall	C recall	Change A to C
buffalo (56)	0.964	0.964	0.964	none
elephant (56)	0.982	0.982	1.000	+1 image
leopard (27)	0.926	0.889	0.963	+1 image
lion (46)	1.000	1.000	1.000	none
rhino (56)	1.000	1.000	0.982	-1 image
zebra (58)	0.983	0.983	0.983	none

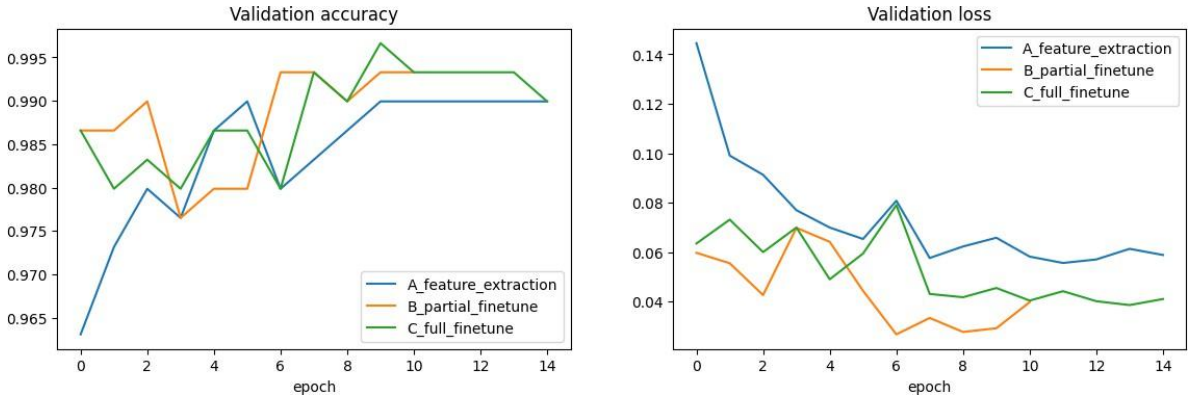


Figure 2. Validation accuracy (left) and validation loss (right) per epoch. B and C epochs are the fine-tuning stage only.

What shifted, and how much

- **A to B:** one more error (-0.33 points accuracy), driven by leopard recall dropping from 25/27 to 24/27. Partial fine-tuning had the lowest validation loss (0.0268) yet the lowest test accuracy, which shows that ranking at this scale is unstable.
- **A to C:** one fewer error (+0.34 points accuracy, +0.51 points macro-F1). Gains came from elephants and leopard, with a loss on rhino.
- **Training dynamics:** A is regularized and capacity-limited (final train loss 0.066 vs validation loss 0.059). B and C reach about 99.7% to 99.9% training accuracy with training loss near 0.004, against validation loss near 0.04. That is a roughly tenfold gap, a sign of mild overfitting that did not hurt validation accuracy.
- **Interpretation:** a frozen ImageNet backbone already separates these six classes at about 98%. Unfreezing layers adds capacity that the remaining errors do not appear to need (see section 4).

4. Error analysis

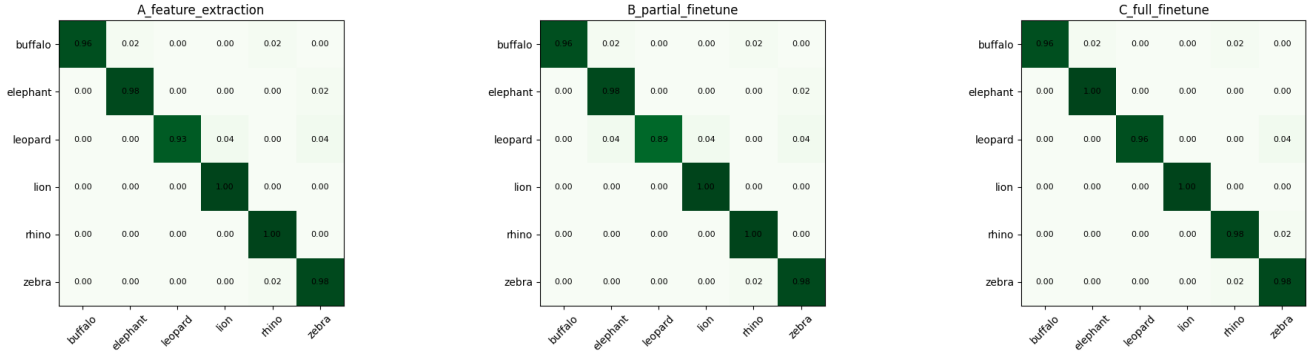


Figure 3. Row-normalized confusion matrices on the test set. Off-diagonal cells are single images (0.02 is 1 of 56; 0.04 is 1 of 27).

4.1 Error pattern across models

Error cell (true to predicted)	A	B	C
buffalo to elephant	1	1	1
buffalo to rhino	1	1	1
zebra to rhino	1	1	1
leopard to zebra	1	1	1
elephant to zebra	1	1	0
leopard to lion	1	1	0
leopard to elephant	0	1	0
rhino to zebra	0	0	1
Total	**6**	**7**	**5**

Four cells appear in every model. The counts match, but the notebook did not save per-image predictions, so it is not verified that the same images are involved each time. The leopard-to-lion confusion (a cat-to-cat error) appears only in A and B and disappears under full fine-tuning.

4.2 The five errors of the best model (C)



Figure 4. All five misclassified test images for Experiment C (true label and prediction above each image).

#	True to predicted	Observed edge case	Likely cause (hypothesis)
1	buffalo to elephant	Herd of buffalo with one large elephant at the right	**Multi-subject scene.** The label is the majority class by box count, but the elephant dominates by area.
2	leopard to zebra	Small animal crossing a dirt road, a vehicle dominates the frame	**Small subject plus context.** The animal's slender build also resembles a cheetah; the Kaggle set has a separate Cheetah class, so the label should be checked by hand.
3	buffalo to rhino	Two close-up buffalo, dark and bulky, little visible horn shape	**Look-alike silhouette.** Large, dark, grass-background quadrupeds are similar across buffalo and rhino.
4	zebra to rhino	Image shows a rhino close-up; no zebra is visible	**Probable label noise.** The prediction is arguably correct.
5	rhino to zebra	Clear rhino, but strongly saturated, HDR-like style, a watermark, and a small second animal in the distance	**Style shift or secondary subject.** This is the only error that looks like a genuine model failure; it is new in C.

4.3 Localized edge-case categories

- **Multi-subject scenes** (errors 1 and 5). Labeling by box count conflicts with what a viewer sees as the main subject. This is a labeling-rule issue as much as a model issue.
- **Small or contextually dominated subjects** (error 2). Vehicles, roads, and wide framing pull attention away from the animal. Leopard is also the rarest class (177 images, 27 in test), so each miss costs 3.7 points of recall.
- **Look-alike silhouettes** (error 3). Bulky dark herbivores are confusable at 224 by 224 resolution.

- **Label noise** (error 4). One of five errors of the best model is plausibly not a model error, so the true ceiling on this test set is above 98.3%.
- **Closed-set limitation** (error 2). The model must choose one of six classes. Look-alike species such as cheetah (present in the Kaggle set but excluded here) will be forced into a class with no way to say, 'none of these'.

5. Engineering conclusions

- **For a low-resource baseline, Experiment A is the defensible default.** It is statistically indistinguishable from C on this test set, trains in minutes, and has 12,294 trainable parameters.
- **Full fine-tuning is worth it only if leopard recall matters** and a larger test set confirms the gain. The evidence here is one image.
- **Partial fine-tuning showed no advantage** in this run, despite the best validation loss.
- **Fix labels and data before adding capacity.** Three of the five errors of the best model trace to labeling rules, label noise, or scene composition.

6. Limitations and threats to validity

- **Species-source confound.** Lion and leopard appear only in the Kaggle data; buffalo, elephant, rhino and zebra only in Ultralytics. Lion's perfect scores in C may partly reflect source cues (framing, resolution, tone). Mixed evidence: leopard errors in A and B went to elephant and zebra as well as lion, so the model is not purely keying on source, but the notebook has no test that separates the two.
- **Possible leakage.** Images were pooled and re-split. If either dataset contains near-duplicate frames or augmented copies, they may span train and test and inflate scores. Not checked.
- **Model selection on the test set.** The 'best' model was chosen by test macro-F1, which gives an optimistic estimate for C. Validation metrics should drive that choice.
- **Small test set, single seed, single split.** 299 images, 27 for leopard, no repeated runs. Differences of 1 to 2 images should not be reported as improvements.
- **Label noise from detection-to-classification conversion.** Majority-class-by-box-count labeling is lossy for multi-animal images.
- **No per-image logging.** Error overlaps across models is inferred from confusion-matrix cells only.

7. Recommended next steps

Priority	Action	Question it answers
1	Add Kaggle zebra, elephant and rhinoceros images (all in its class list) so every species has two sources; then evaluate holding out one source.	Is the model recognizing species or source?
2	Relabel multi-animal images by largest box area (or drop them) and rerun.	How much of the error is labeling rule?
3	Audit duplicates with perceptual hashing across the splits; review the zebra-labeled rhino image and the leopard image by hand.	Is there leakage or label noise?
4	Add an 'other' class (e.g. cheetah, tiger, lynx) or a confidence threshold with abstention.	How does it behave on look-alike species?